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Data Science

Chapter 7: Model Evaluation

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- So far: **parameter** and **interval estimation** for a parameter θ of a **probabilistic model** or **distribution** from an empirical sample
- Now: Making a justified decision about the *validity* or *invalidity* of a **hypothesis** about the parameters of a **probabilistic model** or distribution based on available observations (= sample(s)): **parameter testing**
- Parameter tests are a subset of **statistical tests**. Statistical testing means using the information gained (computed) from a sample to make a decision about a **hypothesis**: **hypothesis test**
- A **hypothesis** is an (unproven) assumption about:
 - the distribution, or
 - specific parameters of a population.
- Hypotheses** may arise from:
 - previous statistical observations,
 - theoretical considerations, or
 - informed conjectures.
- Hypothesis tests do *not* decide or *cannot* prove whether the hypothesis is *false* or *true*. As a result of the hypothesis test, the hypothesis is either **retained** („accepted“) or **rejected** („discarded“).
- Important: Control the probabilities of errors:
 - Rejecting a true hypothesis,
 - Retaining a false hypothesis.

Null and alternative hypotheses I

The **null hypothesis** or **initial hypothesis** H_0 is an assumption about parameters θ of a distribution, shown here for a numerical value of a parameter, for example

$$H_0 : \theta \leq \theta_0 \quad \text{oder} \quad H_0 : \theta = \theta_0 \quad \text{oder} \quad H_0 : \theta \geq \theta_0$$

The null hypothesis is regarded as correct until *sufficient evidence* to the contrary has been provided.

The **alternative hypothesis** H_1 is a complementary assumption to H_0 :

$$H_1 = \overline{H_0}$$

Two-sided questions test whether the parameter of interest θ does not have the value θ_0 . The hypotheses are

$$H_0 : \theta = \theta_0 \quad \text{against} \quad H_1 : \theta \neq \theta_0 \quad \text{„two-sided test“}$$

One-sided questions test whether the parameter of interest θ is smaller or larger than the value θ_0 , and the possible hypotheses are

$$H_0 : \theta \leq \theta_0 \quad \text{against} \quad H_1 : \theta > \theta_0 \quad \text{„upper-tailed test“}$$

$$H_0 : \theta \geq \theta_0 \quad \text{against} \quad H_1 : \theta < \theta_0 \quad \text{„lower-tailed test“}$$

Null and alternative hypotheses II

When formulating hypotheses, the following must be strictly observed:

- The **null hypothesis** and all other logically possible statements must be mutually exclusive, i.e. the **null hypothesis** must be formulated so that there is only one other statement: the **alternative hypothesis**.
- The **null hypothesis** should **not** be the statement that is to be **confirmed** by the test procedure and the sample result, but rather the **logical opposite statement**.

Example: A political scientist might begin an empirical study with the null hypothesis that voter support for a chancellor candidate is the same among men and women.

This is due to the **falsification principle**:

- Refuting a statement generally yields more valuable information than confirming it.
- If a statement is confirmed by the sample result, this does not mean that it is correct, but only that the observed data are not sufficient to refute it.

Therefore, instead of speaking of accepting the **null hypothesis**, it is better to speak of „retaining“ or even „not rejecting“ it.

Type 1 and type 2 errors

When testing hypotheses, two errors can occur:

	H_0 accept	H_0 reject
H_0 is true	correct decision	type 1 error
H_0 is false	type 2 error	correct decision

- ① **type 1 error**: The null hypothesis is rejected even though H_0 is true. The probability of type 1 error is

$$P(H_0 \text{ reject} \mid H_0 \text{ is true}) = \alpha$$

- ② **type 2 error**: The null hypothesis is not rejected even though H_0 is false. The probability of type 2 error is

$$P(H_0 \text{ accept} \mid H_0 \text{ is false}) = \beta$$

Usually, the type 1 error is specified, because one fears the risk of type 1 error more; typical values are $\alpha = 5\%$ and $\alpha = 1\%$.

The type 2 error must then be computed from this specification (usually very complicated), because $\beta = 1 - \alpha$ does *not* hold!

Decision I

After the hypotheses H_0 and H_1 have been formulated, the distribution $F_{\Xi}(\xi)$ of a suitable **test statistic** Ξ is derived under the **condition that the null hypothesis is true**.

Then **disjoint acceptance and rejection regions** (usually intervals) are defined so that the **type 1 error is at most α** :

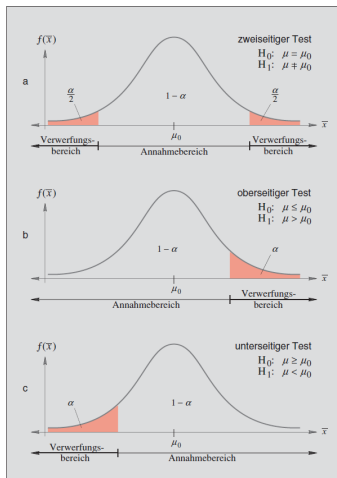
$$P(\xi \in \text{reject region} \mid H_0 \text{ is true}) = \alpha$$

The probability of a type 1 error, α , is also called the **significance level** of the test.

Next, the **concrete test statistic** ξ (as a realization of Ξ) is computed from the sample and checked to see whether it falls into the acceptance or rejection region. **If it falls into the acceptance region, the null hypothesis H_0 is retained; otherwise, it is rejected in favor of H_1 .**

Even though the theoretical background of hypothesis tests is anything but trivial, their practical application is comparatively simple ☺.

Acceptance and Rejection region for two-sided and one-sided hypothesis tests
[Schira, Statistische Methoden der VWL und BWL, 2021]:



Legend: two-sided test = zweiseitiger Test; upper-sided test = oberseitiger Test; lower-sided test = unterseitiger Test, acceptance region = Annahmebereich; rejection region = Verwerfungsbereich

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Two-sided test

Hypotheses: $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$

Acceptance region: $W_0 = [\xi_{c,u}, \xi_{c,o}]$

Rejection regions: $W_1 = (-\infty, \xi_{c,u}) \cup (\xi_{c,o}, \infty)$

The corresponding probabilities are

$$P(\Xi \in W_1) = \underbrace{P(\Xi < \xi_{c,u})}_{\stackrel{!}{=}\alpha/2} + \underbrace{P(\Xi > \xi_{c,o})}_{\stackrel{!}{=}\alpha/2} = \underbrace{F_{\Xi}(\xi_{c,u})}_{=\alpha/2} + \underbrace{1 - F_{\Xi}(\xi_{c,o})}_{=\alpha/2} = \alpha$$

$$P(\Xi \in W_0) = P(\xi_{c,u} \leq \Xi \leq \xi_{c,o}) = F_{\Xi}(\xi_{c,o}) - F_{\Xi}(\xi_{c,u}) = 1 - \alpha$$

where the **significance level** α is split symmetrically. The values $\xi_{c,u}$ and $\xi_{c,o}$ are called the „critical lower“ and „critical upper“ values, respectively, and they follow from the **$\alpha/2$ and $(1 - \alpha/2)$ quantiles** of the distribution $F_{\Xi}(\xi)$ as

$$\xi_{c,u} = \xi_{[\alpha/2]} \text{ and } \xi_{c,o} = \xi_{[1-\alpha/2]}$$

Decision rule:

$$\xi < \xi_{[\alpha/2]} \text{ or } \xi > \xi_{[1-\alpha/2]} \implies \text{reject } H_0!$$

Decision rule (*symmetric* distribution $F_{\Xi}(\xi)$):

$$|\xi| > \xi_{[1-\alpha/2]} \implies \text{reject } H_0!$$

Example: two-sided test

Problem:

In a student pub, beer glasses are supposed to contain $0.4l$.

A sample of size $n = 50$ yields:

- Sample mean: $\bar{x} = 0.38l$
- Sample variance: $s^2 = 0.0064l^2$

Step 1: Hypotheses

$$H_0 : \mu = 0.4l \quad \text{vs.} \quad H_1 : \mu \neq 0.4l$$

Step 2: Standard Deviation

$$\hat{\sigma}_{\bar{X}} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{0.0064l^2}{50}} \approx 0.01143l$$

Step 3: Test Statistic

$$z = \frac{\bar{x} - \mu_0}{\hat{\sigma}_{\bar{X}}} = \frac{0.38 - 0.4}{0.01143} \approx -1.75$$

Step 4: Significance Level ($\alpha = 0.05$) & Critical Value

$$z_{[1-0.025]} = z_{[0.975]} = 1.96$$

Step 5: Decision

$$|z| = 1.75 < 1.96 \quad \implies \quad \text{retain or do not reject } H_0$$

Conclusion:

There is no significant evidence that the mean filling amount differs from $0.4l$.

Upper-sided test

Hypotheses: $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$

Acceptance region: $W_0 = (-\infty, \xi_c]$

Rejection region: $W_1 = (\xi_c, \infty)$

The corresponding probabilities are

$$P(\Xi \in W_1) = \underbrace{P(\Xi > \xi_c)}_{=\alpha} = \underbrace{1 - F_{\Xi}(\xi_c)}_{=\alpha} = \alpha$$

$$P(\Xi \in W_0) = P(\Xi \leq \xi_c) = F_{\Xi}(\xi_c) = 1 - \alpha$$

where the **significance level** α is placed on the *right* (= upper) side. The value ξ_c is called the „critical“ value, and it follows from the **(1 - α) quantile** of the distribution $F_{\Xi}(\xi)$ as

$$\xi_c = \xi_{[1-\alpha]}$$

Decision rule:

$$\xi > \xi_{[1-\alpha]} \implies \text{reject } H_0!$$

Example: upper-sided test

Problem:

A large trucking company pays the traffic fines its 5360 drivers on German highways and has allocated a budget of 1000 € per driver/year. A random sample of $n = 50$ drivers reveals that the average cost per driver last year was $\bar{x} = 1020$ €. Based on many years of experience, the standard deviation has remained fairly constant at $\sigma = 60$ €.

Step 1: Hypotheses

$$H_0 : \mu \leq 1000 \text{ €} \quad \text{vs.} \quad H_1 : \mu > 1000 \text{ €}$$

Step 2: Standard Deviation

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{60 \text{ €}}{\sqrt{50}} \approx 8.485 \text{ €}$$

Step 3: Test Statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma_{\bar{X}}} = \frac{1020 \text{ €} - 1000 \text{ €}}{8.49 \text{ €}} \approx 2.3571$$

Step 4: Significance Level ($\alpha = 0.05$) & Critical Value

$$z_{[1-0.05]} = z_{[0.95]} = 1.645$$

Step 5: Decision

$$z = 2.3571 > 1.645 \quad \implies \quad \text{reject } H_0$$

Conclusion:

The average cost per driver is significantly above 1000 €.

Hypotheses: $H_0 : \theta \geq \theta_0$ against $H_1 : \theta < \theta_0$

Acceptance region: $W_0 = [\xi_c, \infty)$

Rejection region: $W_1 = (-\infty, \xi_c)$

The corresponding probabilities are

$$P(\Xi \in W_1) = \underbrace{P(\Xi \leq \xi_c)}_{=\alpha} = \underbrace{F_{\Xi}(\xi_c)}_{=\alpha} = \alpha$$

$$P(\Xi \in W_0) = P(\Xi > \xi_c) = 1 - F_{\Xi}(\xi_c) = 1 - \alpha$$

where the **significance level** α is placed on the *left* (= lower) side. The value ξ_c is called the „critical“ value, and it follows from the **α quantile** of the distribution $F_{\Xi}(\xi)$ as

$$\xi_c = \xi_{[\alpha]}$$

Decision rule:

$$\xi < \xi_{[\alpha]} \implies \text{reject } H_0!$$

Example: lower-sided test

Problem:

In the previous example w.r.t. beer glasses, a *one-sided test* would be more appropriate, since no one is going to complain if there is too much beer in the glass. The null hypothesis would therefore need to be modified.

Step 1: Hypotheses

$$H_0 : \mu \geq 0.4l \quad \text{vs.} \quad H_1 : \mu < 0.4l$$

Step 2: Standard Deviation

$$\hat{\sigma}_{\bar{X}} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{0.0064l^2}{50}} \approx 0.01143l$$

Step 3: Test Statistic

$$z = \frac{0.38 - 0.4}{0.01143} \approx -1.75$$

Step 4: Significance Level ($\alpha = 0.05$) & Critical Value

$$z_{[0.05]} = -1.645$$

Step 5: Decision

$$z < -1.645 \implies \text{reject } H_0$$

Conclusion:

Evidence that the mean filling is **below** 0.4l.

Test for equality

The known principle for two-sided and one-sided testing a parameter θ can directly be transferred:

Two different, independently drawn samples with sizes n_1 and n_2 are considered. The **test for equality** checks whether the two populations have the same numerical value of a certain distribution parameter.

Two-sided questions test whether the two parameters in question, θ_1 and θ_2 , are equal:

$$H_0 : \theta_1 = \theta_2 \quad \text{against} \quad H_1 : \theta_1 \neq \theta_2 \quad \text{„two-sided test“}$$

One-sided questions test whether the parameter in question θ_1 is smaller or greater than θ_2 . The possible hypotheses are

$$H_0 : \theta_1 \leq \theta_2 \quad \text{against} \quad H_1 : \theta_1 > \theta_2 \quad \text{„upper-tailed test“}$$

$$H_0 : \theta_1 \geq \theta_2 \quad \text{against} \quad H_1 : \theta_1 < \theta_2 \quad \text{„lower-tailed test“}$$

Strictly speaking, $H_0 : \theta_1 = \theta_2$, since $\theta_1 = \theta_2$ is assumed for the test distribution; therefore, α may also be smaller (conservative estimate).

The procedure for making the decision is analogous to the known principle for testing a parameter θ .

Example: test for equality

Problem:

Lamps from two companies X and Y are being tested for their lifetime. Samples are drawn with $n_X = 64$, $\bar{x} = 1512\text{h}$, $s_X = 12\text{h}$ and $n_Y = 58$, $\bar{y} = 1504\text{h}$, $s_Y = 15\text{h}$. May one conclude that company X supplies lamps with a longer lifetime?

Step 1: Hypotheses

$$H_0 : \mu_X \leq \mu_Y \quad \text{vs.} \quad H_1 : \mu_X > \mu_Y$$

Step 2: Standard Deviation

$$\hat{\sigma}_\Delta = \sqrt{\frac{n_X \cdot s_X^2 + n_Y \cdot s_Y^2}{n_X + n_Y - 2}} \cdot \sqrt{\frac{n_X + n_Y}{n_X \cdot n_Y}} \approx 2.47$$

Step 3: Test Statistic

$$t = \frac{\bar{x} - \bar{y}}{\hat{\sigma}_\Delta} = \frac{1512 - 1504}{2.47} \approx 3.24$$

Step 4: Significance Level ($\alpha = 0.025$) & Critical Value

$$t_{n_X + n_Y - 2, [1-\alpha]} = t_{120, 0.975} \approx 1.98 \quad (\text{Student-t-distribution})$$

Step 5: Decision

$$t = 3.24 > 1.98 \quad \implies \quad \text{reject } H_0$$

Conclusion:

Company X produces lamps with significantly longer lifetimes.

Summary & illustration

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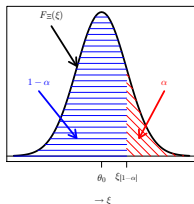
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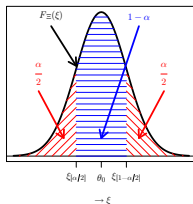
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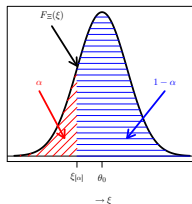
Oberseitiger Test



Zweiseitiger Test



Unterseitiger Test



The formal procedure for carrying out a hypothesis test always consists of the following steps:

- 1 Formulate the hypotheses H_0 and H_1
- 2 From the sample: compute
 - a) required values (based on point estimates)
 - b) test statistic ξ
- 3 Specify α and determine the critical value(s) (quantile(s) of the test distribution $F_{\Xi}(\xi)$)
- 4 Make the test decision by comparing the critical value(s) and the test statistic ξ

P-value & significance I

- So far: Specification of α and no consideration of the distance between the computed test statistic ξ and the critical value relevant for the test decision, which is defined by α .
- Now: Specification of the probability $\tilde{\alpha}$ with which the computed test statistic ξ , under a true null hypothesis, would lie exactly on the boundary(ies) between acceptance and rejection region(s): **P-value**. The smaller this value, the more **significant** the rejection of the null hypothesis is, i.e. the more willing one is to accept the alternative hypothesis.

Definition of $\tilde{\alpha}$ (P-value)

- Upper-sided test:

$$\tilde{\alpha} = P(\Xi \geq \xi) = 1 - F_{\Xi}(\xi)$$

- Lower-sided test:

$$\tilde{\alpha} = P(\Xi \leq \xi) = F_{\Xi}(\xi)$$

- Two-sided test:

$$\tilde{\alpha} = 2 \cdot \min \{P(\Xi \leq \xi), P(\Xi \geq \xi)\}$$

Type	$F_{\Xi}(\xi)$	α	$\xi_{c,u}$	$\xi_{c,o}$	ξ	$\tilde{\alpha} = P\text{-value}$
two-sided	SND	0,05	-1,960	1,96	0,00	1,000
two-sided	SND	0,05	-1,960	1,96	0,05	0,960
two-sided	SND	0,05	-1,960	1,96	1,00	0,317
two-sided	SND	0,05	-1,960	1,96	1,96	0,050
two-sided	SND	0,05	-1,960	1,96	3,00	0,003
lower-sided	SND	0,05	-1,645	—	3,00	0,999
lower-sided	SND	0,05	-1,645	—	0,00	0,500
lower-sided	SND	0,05	-1,645	—	-1,00	0,159
lower-sided	SND	0,05	-1,645	—	-1,64	0,050
lower-sided	SND	0,05	-1,645	—	-3,00	0,001
upper-sided	χ^2_{29}	0,10	—	39,1	5,00	0,999
upper-sided	χ^2_{29}	0,10	—	39,1	30,00	0,414
upper-sided	χ^2_{29}	0,10	—	39,1	39,00	0,102
upper-sided	χ^2_{29}	0,10	—	39,1	39,10	0,100
upper-sided	χ^2_{29}	0,10	—	39,1	50,00	0,009

Usually, α is specified before the test, and the null hypothesis is then rejected if $\tilde{\alpha}$ is less than or equal to α . The smaller $\tilde{\alpha}$ is compared to α , the more **significant** the rejection of H_0 is!